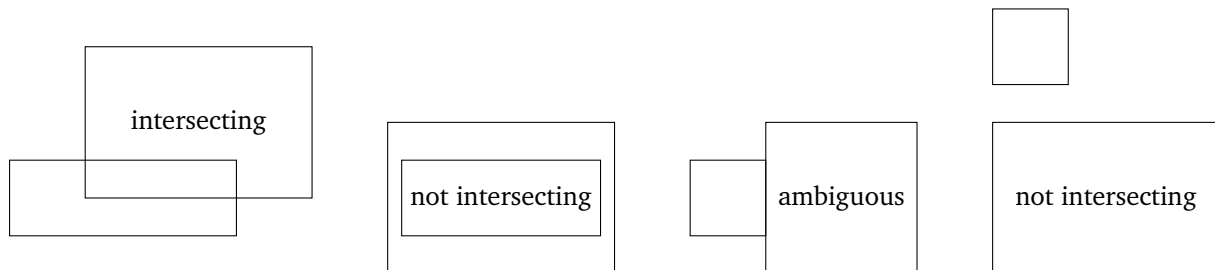


Your mission, should you choose to accept, is to write a program that calculates whether or not two shapes are intersecting. Intersection, for the sake of definition, is defined as one or more (but not all) of the set of points belonging to one figure being contained inside of another.

Visual examples of what does and does not count can be found below.



Although this may appear to be a somewhat trivial assignment, applicants are welcomed to extend the problem definition however they see fit. Please flex as hard as possible.

Examples of extensions:

- Extending the dimensionality of the intersecting figures. IE, using 3d polyhedra instead of 2d polygons.
- Creating a web API to determine whether or not figures intersect.
- Get creative.

Submissions are encouraged to be done in any commonly used programming language. C, C++, C#, Java, Python, Go, Rust, Ruby, and flavors of Javascript are all acceptable. Please provide a README with your submission that details language versions, package versions (if not already included in a package.json or requirements.txt, for example), and instructions for running your software with valid and invalid test cases.